ASSIGNMENT NO. 1

Course: SE-458 – Operating Systems

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Task 1:

Understanding of Operating System Structure

Introduction

An Operating System (OS) is one of the most important software components of a computer. It acts as an interface between the user and the computer hardware. Without an operating system, users cannot communicate directly with the hardware or run application programs efficiently. The operating system manages all hardware resources, provides services to software, and ensures that different programs can run smoothly without interfering with each other.

Evolution and Key Features

The operating system has evolved over many generations. In the early days, computers executed only one program at a time. As computer technology im-proved, operating systems introduced features such as multiprogramming, multitasking, time-sharing, memory management, and process scheduling. These improvements allowed computers to utilize the CPU more efficiently and support multiple users and applications.

One of the most important concepts discussed in the document is **system calls**. A system call provides a controlled way for user programs to request services from the operating system. Since application programs run in user mode, they cannot directly access hardware resources. Whenever a program needs to per-form operations such as reading a file, writing data, allocating memory, or communicating with hardware devices, it requests the operating system through a system call. The CPU temporarily switches from user mode to kernel mode, per-forms the requested service, and then returns control to the user program. This mechanism improves both security and stability of the computer system.

Hardware Management

The operating system also manages important hardware resources including:

- Processor (CPU)
- Main Memory (RAM)
- Cache Memory
- Secondary Storage (Hard Disk/SSD)
- Input and Output Devices
- System Bus
- Device Controllers
- Device Drivers

Each hardware component has a specific role, and the operating system coordinates their communication so that programs execute correctly and efficiently.

Important Points from Hardware Components

1. Processor (CPU)

The CPU is known as the brain of the computer. It executes instructions using the Fetch – Decode – Execute Cycle. It also contains registers such as the Program Counter (PC), Stack Pointer (SP), and Program Status Word (PSW), which help control program execution. Modern processors also use pipelining and superscalar architecture to improve performance.

2. Memory

The operating system manages different types of memory arranged in a hierar-chy. The memory hierarchy consists of Registers, Cache Memory, Main Memory (RAM), Hard Disk, and Magnetic Tape. The higher levels are faster but smaller and more expensive, while the lower levels are slower but provide much larger storage capacity.

3. Input and Output Devices

Input and output devices communicate with the operating system through de-vice controllers and device drivers. The controller directly controls the hard-ware device, while the device driver is software that allows the operating system to communicate with that controller.

4. System Bus

The system bus is the communication pathway that connects the CPU, memory, and I/O devices. It allows data, addresses, and control signals to move between different hardware components.

5. Memory Management Unit (MMU)

The MMU converts virtual addresses into physical addresses and provides mem-ory protection between programs.

6. Device Drivers

A device driver is special software that allows the operating system to communi-cate with hardware devices. Different hardware devices require different drivers to function correctly.

Summary

After studying the Operating System Structure document, I understood that an operating system is much more than software that starts a computer. It man-ages hardware resources, provides services to applications through system calls, controls memory, handles input and output devices, and ensures that multiple programs run safely and efficiently.

Question 1

Explain the role of the mainboard (motherboard) in a computer system. List and describe at least four major components commonly found on the mainboard and their functions.

Introduction

The motherboard, also known as the mainboard, is the primary printed circuit board of a computer. It serves as the central communications hub, connecting all the hardware components and allowing them to interact with each other efficiently.

Role and Explanation

Every major component of the computer, including the processor, memory, stor-age devices, graphics card, and input/output devices, is connected to the motherboard either directly or through expansion slots. The motherboard distributes power to all connected components and provides communication pathways through electrical traces known as buses. Without a motherboard, individual hardware components cannot work together as one complete computer system.

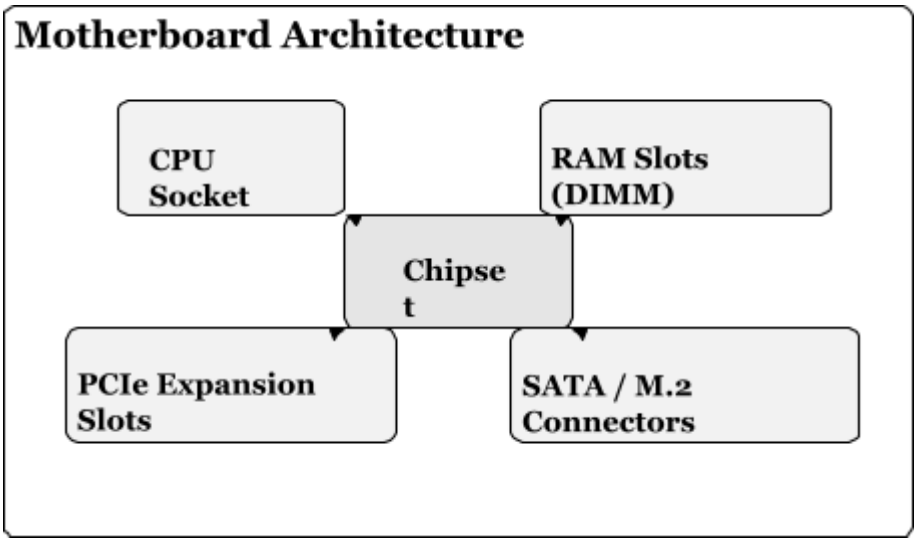


Figure 1: Logical Layout of Major Motherboard Components

Major Components of a Motherboard

1. **CPU Socket:**

   The CPU socket is the place where the processor is installed. It holds the processor securely, provides electrical connections between the CPU and motherboard, and facilitates communication between the proces-sor and other hardware.

2. **RAM Slots (DIMM Slots):**

   RAM slots are used to install main memory mod-ules. They store data and programs currently being used, provide fast tem-porary memory for the CPU, and allow easy memory upgrades.

3. **Chipset:**

   The chipset manages communication between the CPU, RAM, stor-age devices, and input/output devices. It controls data flow, supports USB, SATA, PCIe, and other interfaces, and dictates overall system capabilities.

4. **Expansion Slots (PCI Express):**

   Expansion slots allow users to install addi-tional hardware components such as graphics cards, sound cards, network cards, and Wi-Fi adapters. They increase computer functionality and sup-port hardware upgrades.

5. **BIOS/UEFI Chip:**

   The firmware is stored on a special memory chip. It starts the computer during booting, checks whether hardware components are working properly (POST), and loads the operating system into memory.

Final Remarks

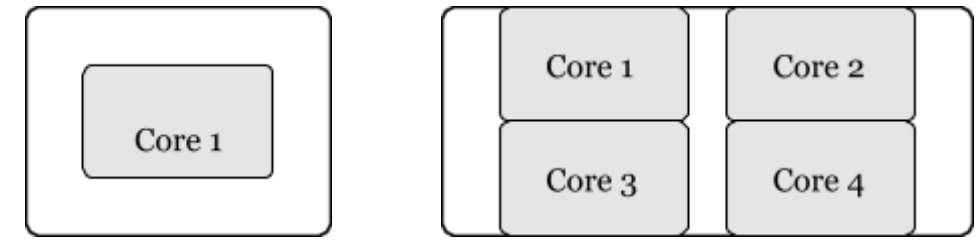
The motherboard acts as the backbone of the computer system. Its components, such as the CPU socket, RAM slots, and chipset, work in harmony to ensure smooth communication and efficient performance across the entire machine.

Question 2

Differentiate between single-core and multi-core processors. What are the advantages of multi-core processors in terms of system performance and multitasking?

Introduction

A processor (CPU) is the main component of a computer that executes instruc-tions and performs calculations. A core is an independent processing unit capable of fetching, decoding, and executing instructions. Depending on the architecture, processors can feature either one core or multiple cores on a single chip.



Single-Core Processor **Multi-Core Processor (Quad-Core)**

Figure 2: Single-Core vs. Multi-Core Processor Architecture

Difference between Single-Core and Multi-Core Proces-sors

Feature	Single-Core Processor	Multi-Core Processor
Cores	Has only one processing core.	Has two or more processing cores.
Execution	Can execute only one task at a time.	Can execute multiple tasks simultaneously.
Multitasking	Slower while multitasking.	Faster and more efficient in multitasking.
Performance	Lower overall perfor-mance.	Higher overall perfor-mance.
Use Case	Suitable for simple applica-tions.	Suitable for modern soft-ware, gaming, and profes-sional work.
Workload Management	Higher chance of CPU becoming overloaded.	Workload is shared among multiple cores, reducing overload.

Table 1: Comparison of Processor Types

Advantages of Multi-Core Processors

- **Better Multitasking:** A multi-core processor can run multiple programs at the same time without significantly slowing down the computer.
- **Higher Performance:** Since multiple cores work together in parallel, programs can complete complex computational tasks much faster.
- **Improved System Responsiveness:** The operating system can distribute background processes and active foreground applications among different

cores, reducing system lag.

- **Efficient Resource Utilization:** The operating system's scheduler distributes processes intelligently, improving overall CPU utilization.
- **Support for Modern Applications:** Applications such as video editing soft-ware, IDEs, and virtual machines are specifically written (multi-threaded) to utilize multiple cores efficiently.

Summary

While single-core processors paved the way for early computing, multicore processors are essential for modern computer architecture, providing the necessary parallel computing power for seamless multitasking and high-performance applications.

Question 3

Draw and label the memory hierarchy of a com-puter system. Briefly describe the role and rela-tive speed of each level from registers to secondary storage.

Introduction

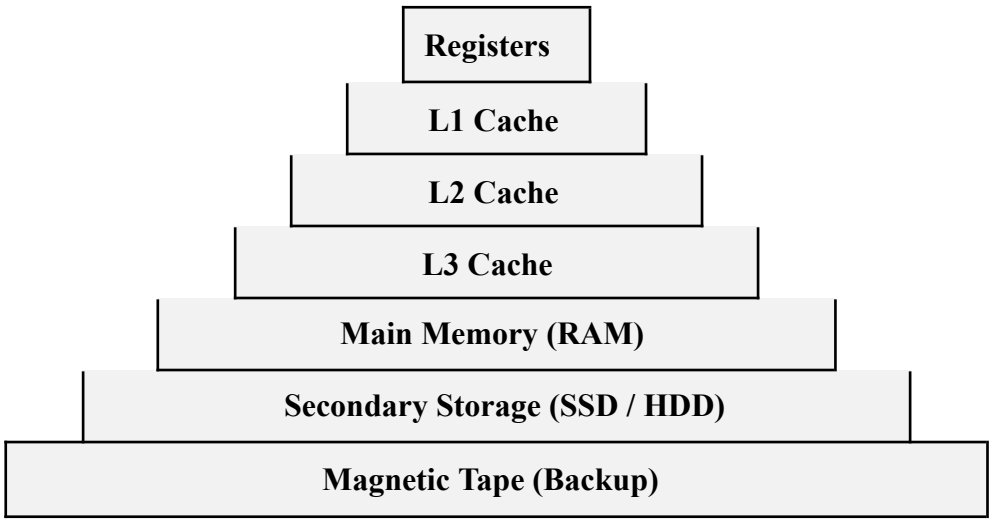


Figure 3: Computer Memory Hierarchy Pyramid

A computer uses different types of memory arranged in a structured hierarchy. The memory hierarchy is designed to provide a delicate balance between speed, storage capacity, and cost. Memory closer to the CPU is faster but smaller and more expensive, while memory farther from the CPU is slower but provides mas-sive storage capacity at a lower cost.

Description of Each Level

1. **Registers:**

   The fastest memory, located directly inside the CPU. They store immediate data, program counters, operands, and temporary values cur-rently being processed. Capacity is measured in bytes.

2. **Cache Memory (L1, L2, L3):**

   High-speed SRAM that stores frequently used data and instructions so the CPU can access them without waiting for RAM. L1 is the smallest and fastest (inside the core), while L3 is larger and shared across cores.

3. **Main Memory (RAM):**

   Volatile memory that temporarily holds active programs and provides working memory for the CPU. It has a large capacity (Gigabytes) and fast speed, but is slower than cache.

4. **Secondary Storage (SSD/HDD):**

   Non-volatile memory used for permanent data storage. It stores the operating system, files, and applications even when the power is turned off.

5. **Magnetic Tape (Backup Storage):**

   Extremely large capacity, very slow sequential access storage used primarily for long-term archiving and disaster recovery.

Summary Table

Memory Level	Relative Speed	Capacity	Volatile
Registers	Fastest	Very Small (Bytes)	Yes
L1 / L2 Cache	Very Fast	Small (MBs)	Yes
L3 Cache	Fast	Medium (MBs)	Yes
Main Memory (RAM)	Medium	Large (GBs)	Yes
Secondary Storage	Slow	Very Large (TBs)	No
Magnetic Tape	Very Slow	Huge (PBs)	No

Table 2: Characteristics of Memory Hierarchy Levels

This hierarchical design optimizes computer performance by keeping frequently used data as close to the CPU as possible, while providing economical mass storage at the lower levels.

Question 4

What are system buses? Identify and describe the three main types of buses in a computer system and explain how they interact with hardware components.

Introduction

A system bus is a communication pathway that connects the CPU, memory, and input/output devices. It serves as a shared electronic highway, allowing different hardware components to exchange information and synchronize operations. Without a system bus, isolated hardware components would be completely unable to communicate.

Types of System Buses

The system bus is technically divided into three distinct sub-buses, each with a specific purpose.

1. **Data Bus:**

   The data bus transfers the actual data payload between the CPU, memory, and input/output devices. The width of this bus (e.g., 64-bit) determines how much data can be transferred in a single cycle.

2. **Address Bus:**

   The address bus carries the memory addresses that specify exactly where data should be read from or written to. Unlike the data bus, it usually operates in one direction (from the CPU to memory or I/O).

3. **Control Bus:**

   The control bus carries control signals that coordinate the activities of the computer. It transmits essential commands such as read signals, write signals, interrupt

requests, and clock signals to keep compo-nents synchronized.

Interaction of Hardware Components

These three buses work in unison whenever the CPU accesses memory or a peripheral device.

Example of a Memory Read Operation:

- The CPU places the memory location it needs to access onto the **Address Bus**.
- The CPU sends a *Memory Read* signal down the **Control Bus**.
- The Main Memory receives the address and the read command, retrieves the data, and places it onto the **Data Bus**.
- The CPU reads the data from the Data Bus into its registers.

Key Points

The system bus is a vital part of computer architecture. The Address Bus pin-points locations, the Data Bus handles the heavy lifting of information transfer, and the Control Bus acts as the traffic controller, ensuring all operations occur in the correct sequence.

Question 5

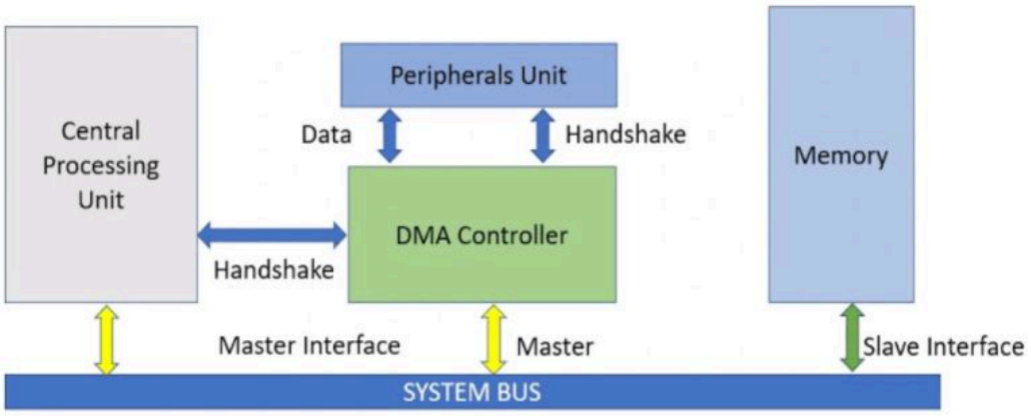
Explain the concept of Direct Memory Access (DMA). How does DMA improve system performance during data transfer between memory and I/O devices?

Introduction

Direct Memory Access (DMA) is a sophisticated hardware technique that allows input/output (I/O) devices to transfer data directly to or from the main memory (RAM) without requiring continuous, manual intervention from the CPU.

The Problem Without DMA

Normally, when data is transferred between an I/O device (like a hard drive) and memory, the CPU controls every single step. It reads a byte from the disk, stores it in a register, and then writes it to RAM. This process, known as Programmed I/O, keeps the processor completely occupied, preventing it from doing other useful work and crippling system performance during large file transfers.



Working of DMA

With DMA, a special hardware module called the DMA Controller (DMAC) takes over.

- The CPU receives a request and configures the DMAC by providing the start-ing memory address, the byte count, and the transfer direction (read or write).
- The CPU then returns to executing other programs.
- The DMAC requests control of the system bus and directly shuttles data between the I/O device and RAM.
- Once the block transfer is complete, the DMAC sends a hardware interrupt to notify the CPU.

Advantages of DMA

1. **Reduces CPU Workload:**

   The processor delegates the tedious task of mov-ing data blocks, freeing it up for computation.

2. **Faster Data Transfer:**

   Specialized DMA hardware transfers massive amounts of data more efficiently than the CPU.

3. **Better Multitasking:**

   The operating system can schedule and run other processes concurrently while the DMAC handles background file copying.

Final Remarks

By offloading routine data movement to a dedicated controller, DMA significantly boosts overall system throughput, making tasks like large file transfers, video streaming, and disk backups seamless and efficient without freezing the operat-ing system.

Question 6

Explain the function of the CPU. Describe how the Fetch–Decode–Execute cycle works with the help of a labelled diagram.

Introduction

The Central Processing Unit (CPU) acts as the brain of the computer. It is respon-sible for executing program instructions, performing arithmetic and logical operations, controlling hardware components, and processing data. Every single instruction executed by a computer must pass through the CPU.

Functions of the CPU

The CPU handles several critical operations:

- Fetches instructions from memory.
- Decodes instructions to understand required actions.
- Executes complex arithmetic calculations (via the ALU).
- Processes logical operations (AND, OR, NOT).
- Coordinates synchronization between memory and external peripherals.

The Fetch–Decode–Execute Cycle

The CPU performs all operations using an endless loop known as the Fetch–Decode–Execute Cycle. As long as the computer is powered on, this cycle repeats millions or billions of times per second

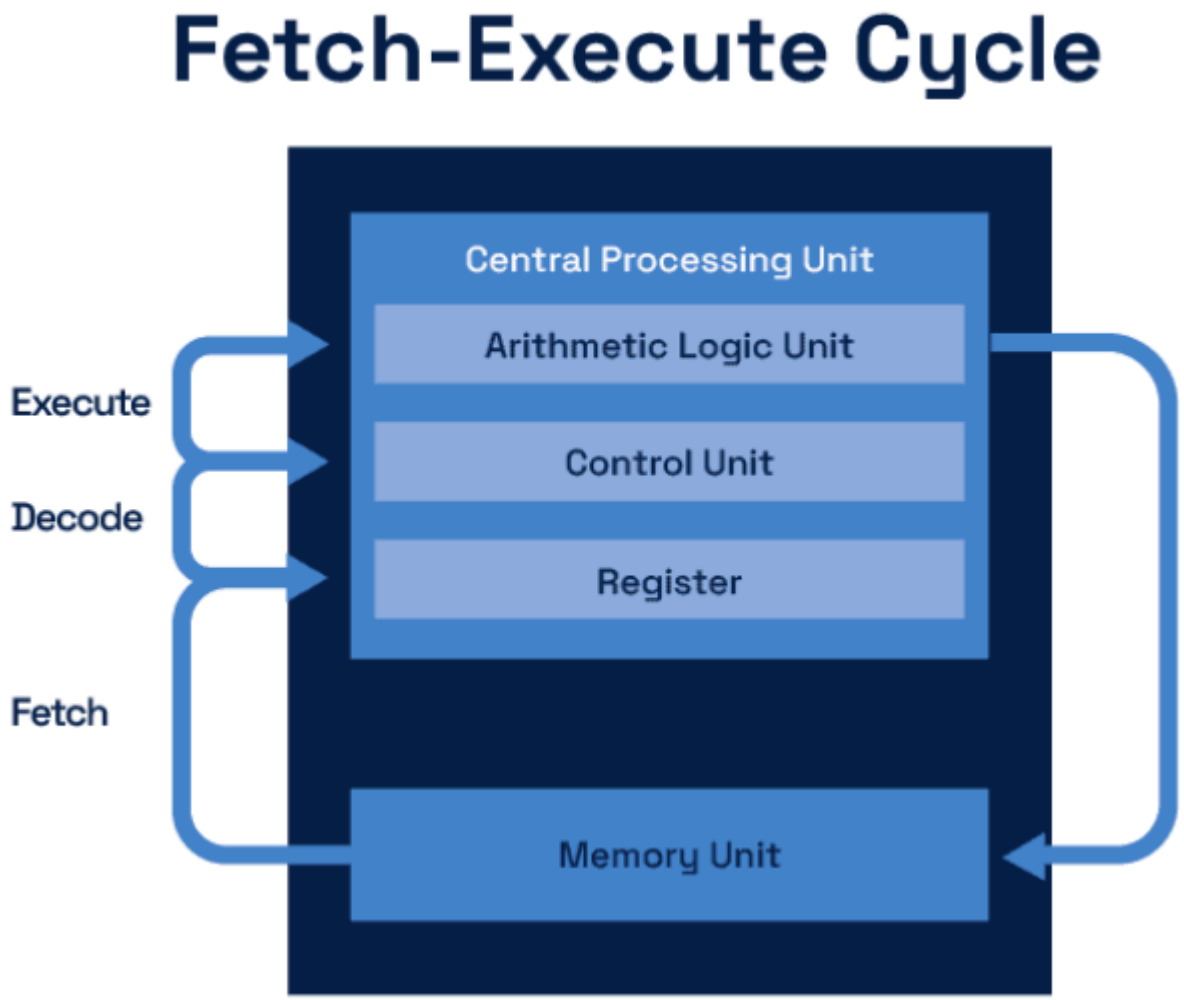


Figure 6: The CPU Fetch-Decode-Execute Cycle

Step 1: Fetch

The CPU retrieves the next instruction from main memory. It uses a specific reg-ister called the Program Counter (PC) to track the memory address of the current instruction.

Step 2: Decode

The Control Unit (CU) receives the fetched instruction and decodes it. It trans-lates the machine code into specific control signals that determine what the processor's hardware needs to do (e.g., add two numbers, move data).

Step 3: Execute

The Arithmetic Logic Unit (ALU) or another internal execution unit performs the required operation. If the instruction was an addition, the ALU calculates the sum. Finally, the result is stored back into a register or main memory, the PC is incremented, and the cycle repeats.

Example

Consider the programming instruction A = B + C. The CPU fetches this command from memory, decodes it to recognize an addition operation, fetches the values of B and C into its registers, executes the addition via the ALU, and finally stores the resulting value A back into memory.

Question 7

Explain the different levels of cache memory (L1, L2, L3) in a CPU. What role does cache memory play in speeding up system performance?

Introduction

Cache memory is an ultra-fast, small-capacity memory hardware built directly into or placed very close to the CPU chip. Its primary purpose is to store frequently used instructions and data, acting as a high-speed buffer between the ultra-fast processor and the relatively slower Main Memory (RAM).

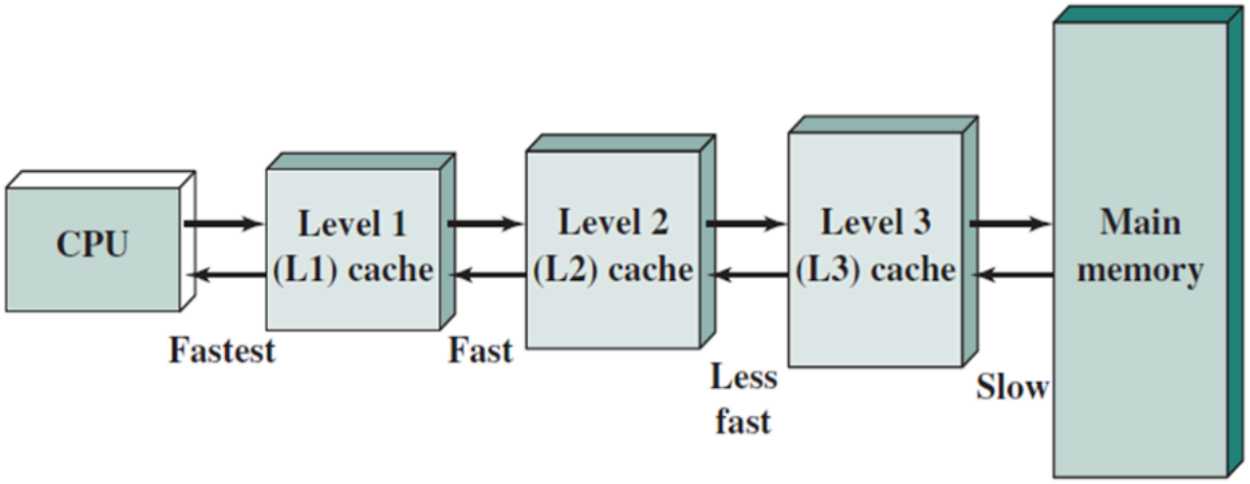


Figure 7: Multi-Level Cache Architecture within a CPU

Levels of Cache Memory

Modern processor architectures implement a multi-tiered caching system to optimize data retrieval.

1. **L1 Cache (Level 1):**

   This is the primary cache embedded directly inside each individual CPU core. It operates at the same speed as the processor, making it the fastest memory available. However, due to space and thermal constraints, it has the smallest capacity (typically 32KB to 128KB).

2. **L2 Cache (Level 2):**

   Slightly slower and larger than L1, the L2 cache serves as the next fallback layer. It is usually dedicated to a single core but is placed slightly further away on the chip. It typically holds hundreds of kilobytes up to a few megabytes.

3. **L3 Cache (Level 3):**

   This is the largest cache level on the processor. Unlike L1 and L2, the L3 cache is shared among all the cores on the CPU. It is slower than L1 and L2 but still drastically faster than fetching data from external RAM.

Role in System Performance

Cache memory fundamentally solves the ”von Neumann bottleneck”—the issue where a fast CPU spends most of its time idling while waiting for data from slow RAM.

- **Spatial and Temporal Locality:** Cache takes advantage of the fact that if a program accesses a specific variable, it is highly likely to access it again soon (temporal locality) or access adjacent variables (spatial locality).
- **Reduced Latency:** By keeping this localized data in SRAM cache, the CPU avoids the high latency of crossing the system bus to query RAM.

Final Remarks

Without cache memory, even a powerful multi-core CPU running at 5 GHz would perform terribly, as it would constantly stall while waiting for main memory data. Cache bridges the massive speed gap between the processor and RAM.

Question 8

Define the function of a system clock in a computer. How does the clock speed (measured in GHz) im-pact the performance of the CPU?

Introduction

The system clock is an internal microchip (often utilizing a quartz crystal oscil-lator) that regulates the timing and speed of all computer functions. It gener-ates a continuous pulse of electrical signals at exact intervals, acting much like a metronome for a musician.

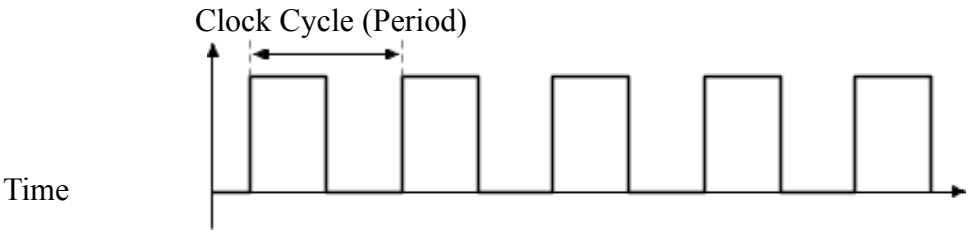


Figure 8: Digital Square Wave Representation of a System Clock Signal

Functions of the System Clock

- **Synchronization:** It ensures that all internal components (CPU, RAM, buses) operate in perfect harmony.
- **Instruction Timing:** Every stage of the Fetch-Decode-Execute cycle requires one or more clock pulses to progress.
- **System Stability:** By maintaining strict timing boundaries, it prevents data collisions on the system bus.

Impact of Clock Speed

Clock speed is measured in Gigahertz (GHz), where 1 GHz equals one billion clock cycles per second. The clock speed dictates exactly how many basic operations the CPU can perform every second.

- **Higher Speed:** A 4.0 GHz processor ticks 4 billion times a second. Gener-ally, a higher clock speed means the processor can execute more instruc-tions in a shorter amount of time, leading to faster program execution, bet-ter responsiveness, and higher framerates in gaming.
- **The IPC Factor:** Clock speed alone is not the sole indicator of performance. Instructions Per Clock (IPC) is equally important. A modern 3.0 GHz CPU might outperform an older 4.0 GHz CPU because modern architectures can execute more instructions during a single clock cycle due to better pipelining and larger caches.

Key Takeaways

While core count defines how many tasks a CPU can handle simultaneously, the clock speed determines how fast it can complete a single task. Both metrics must be balanced for optimal computer performance.